

General instructions: After reading the questions given below, follow the steps described in `Instructions.pdf` to see the input and the expected output for each test case.

1 Monsters

You and some monsters are in a labyrinth. When taking a step to some direction in the labyrinth, each monster may simultaneously take one as well. Your goal is to reach one of the boundary squares without ever sharing a square with a monster.

Your task is to find out if your goal is possible, and if it is, print the length of the shortest such path that you can follow. Your plan has to work in any situation; even if the monsters know your path beforehand.

Input Format:

- The first line of input contains two space separated integers n and m where n is the height and m is the width of the map.
- The next n lines contain m characters each. Each character is either `.` (floor), `#` (wall), `A` (start) or `M` (monster). **NOTE: There is exactly 1 `A` in the input.**

Output Format:

Length of the shortest safe path possible. **If no such path is possible, print -1.**

Constraints:

$$1 \leq n \leq 1000$$

$$1 \leq m \leq 1000$$

Expected Time Complexity:

$$O(n * m)$$

TEST CASE 1**n:** 5**m:** 8
$$\begin{bmatrix} \# & \# & \# & \# & \# & \# & \# & \# \\ \# & M & . & . & A & . & . & \# \\ \# & . & \# & . & M & \# & . & \# \\ \# & M & \# & . & . & \# & . & . \\ \# & . & \# & \# & \# & \# & \# & \# \end{bmatrix}$$
Output: 5**Explanation:**

You can follow this path: Right -> Right -> Down -> Down -> Right

2 Longest Cycle in a Graph

You are given a **directed** graph of n nodes numbered from 0 to $n - 1$, where **each node has at most one outgoing edge**.

The graph is represented with a given 0-indexed array `edges` of size n , indicating that there is a directed edge from node i to node `edges[i]`. If there is no outgoing edge from node i , then `edges[i] = -1`.

Return the length of the longest cycle in the graph. **If no cycle exists, return -1.**

A cycle is a path that starts and ends at the same node.

Input Format:

- The first line of input contains an integer n where n represents the number of nodes in the graph.
- The next line contains n space separated integers representing the values in the `edges` array.

Output Format:

Length of the longest possible cycle. If no cycle exists, return -1.

Constraints:

$$2 \leq n \leq 10^5$$

$$-1 \leq \text{edges}[i] \leq n - 1$$

$$\text{edges}[i] \neq i$$

Expected Time Complexity:

$O(n)$

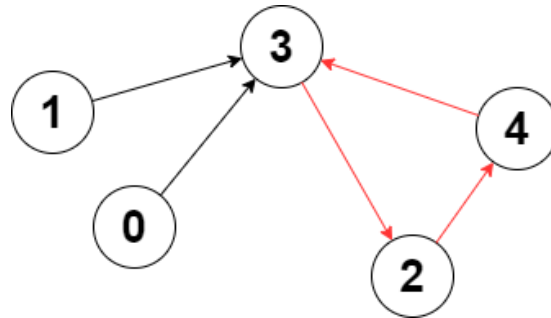


Figure 1: Figure for Test Case 1.

TEST CASE 1

n: 5

edges: [3, 3, 4, 2, 3]

Output: 3

Explanation: The longest cycle in the graph is the cycle: 2 -> 4 -> 3 -> 2. The length of this cycle is 3, so 3 is returned.

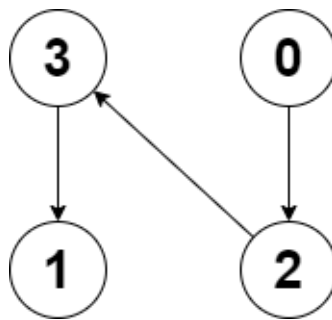


Figure 2: Figure for Test Case 2.

TEST CASE 2

n: 4

edges: [2, -1, 3, 1]

Output: -1

Explanation: There are no cycles in this graph, so -1 is returned.